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Original Article

Does engagement in frequent physical activity improve diabetes mellitus among older adults in India? A propensity score matching approach

Pradeep Kumar^a, Ratna Patel^b, T. Muhammad^b, Shobhit Srivastava^{c,*}^a Population Council, India Country Office, 5A, GF, India Habitat Centre, New Delhi, Delhi, 110003, India^b International Institute for Population Sciences, Mumbai, Maharashtra, 110048, India^c Department of Research and Innovation, Mamta Health Institute for Mother and Child, New Delhi, 110048, India

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ABSTRACT

Background and aims: Diabetes is a major chronic condition and was identified as one of the four priority non-communicable diseases that the United Nations targeted because of its increasing disease burden. The present study aims to examine the impact of physical activity on improving diabetes mellitus among older adults in India.

Methods: This study utilizes data from India's first nationally representative longitudinal ageing survey (2017–18). Descriptive along with bivariate analysis was used to present the preliminary results. Additionally, Propensity score matching (PSM) analysis was used.

Results: About 14% of older adults suffered from diabetes. Eighteen percent of older adults did frequent physical activity. The estimated average treatment effect on the treated (ATT) values in treated and control groups were 0.123 and 0.147, respectively, indicating that the prevalence of diabetes among older adults was reduced by 2.5% points because of frequent physical activity. The average treatment effect on the untreated (ATU) results indicates that among older adults who did not do frequent physical activity, if they did frequent physical activity, the prevalence of diabetes is likely to decrease by 2.2% points.

Conclusion: Our findings show that frequent physical activity is associated with a lower risk of diabetes in older adults. The results underscore the need to develop strategies of promoting an active lifestyle by maintaining physical activity to combat the diabetes epidemic in the older population.

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1. Introduction

Diabetes is a major chronic condition and was identified as one of the four priority non-communicable diseases that the United Nations targeted because of its increasing disease burden [1]. Diabetes mellitus has persisted as a substantial public health concern over the past quarter-century in India and across the globe [1]. The number of people with diabetes has risen from 108 million in 1980 to 422 million in 2014 [2]. In India, the number of diabetic people is expected to rise from 77 million in 2019 to 101 million in 2030 and further to 134 million by 2045 [3]. India is burdened with a significant global diabetes burden and is considered the diabetic

capital of the world [1]. Population ageing is accredited to be one of the predominant factors leading to the rise in diabetes prevalence in India [4]. Several studies have highlighted that the prevalence of diabetes remains higher among older adults than in middle-aged or young adults [5].

An increase in life expectancy has led to an increase in the number of older adults in India [6]. The rise in the number of older adults has been a cause of concern as it has been associated with an increase in the prevalence of chronic diseases, including diabetes [7]. Several other factors have been linked to the risk of diabetes among older adults, including a family history of diabetes [8], gender of the older adult [8], dietary habits [9], BMI [10]. Also, various socioeconomic factors such as wealth index, religion, caste and educational status have been linked to the risk of diabetes among older adults [11]. Not only this, diabetes has been associated as a comorbid condition with several other chronic conditions [12]. Among various risk factors of diabetes [13], obesity is one of the

* Corresponding author.

E-mail addresses: pradeepiips@yahoo.com (P. Kumar), ratnapatelbhu@gmail.com (R. Patel), muhammad.iips@gmail.com (T. Muhammad), shobhitsrivastava889@gmail.com (S. Srivastava).

most prominent factors [14]. Studies have also highlighted the association between physical activity and obesity, where reduced physical activities have been linked to a higher risk of diabetes in the study population [15,16].

How does engagement in physical activity affect diabetes among the older population? Globally, a growing literature has examined this notion and suggested various pathways through which physical activity define the occurrence of diabetes. Physical activity results in increased energy expenditure, thereby altering the metabolic rate to reduce the risk of diabetes [17]. Another pathway defining the role of physical activity in reducing the risk of diabetes states that physical inactivity affects the insulin in the body and thereby has a role in determining the onset of diabetes [18]. Physical activity is linked to bringing improvement in lipid profile, decreased body weight and percentage of body fat, lower blood pressure, thereby reducing the risk of diabetes [19]. Moreover, physical activity has been linked to reducing intra-abdominal fat, which is a well-known risk factor for insulin resistance, thereby reducing the risk of diabetes [20].

Despite well-established causal inferences and pathways examining the association between physical activity and diabetes [21], studies examining this association among older adults utilizing information from large nationally representative datasets in the Indian context have largely remained elusive. A large share of literature in Western countries agreed that physical activity lowers the risk of diabetes and lowers morbidity and mortality among people with diabetes [22]. In contrast, such studies in the Indian context are somewhat limited, and hardly any study exclusively examined the association between physical activity and diabetes among the older population. Tripathy et al. (2017), examining risk factors of diabetes among 2499 people aged 18–69 years old in a North-Indian state, noted physical activity as one of the factors of diabetes [23]. Furthermore, a study from the Global Burden of Disease group in India noted that low physical activity contributed to almost 2% of disability-adjusted life-years (DALYs) due to diabetes [24]. The unavailability of relevant studies examining the link between physical activity and diabetes among older adults prompted us to examine this under-explored research domain in the Indian context utilizing information from the latest release of the Longitudinal Ageing Survey of India (LASI). The present study aims to examine the impact of physical activity on the improvement of diabetes mellitus among older adults in India.

2. Materials and methods

2.1. Data

This study utilizes data from India's first nationally representative longitudinal ageing survey (LASI 2017–18), which investigates the health, economics and social determinants and consequences of population ageing in India [25]. The representative sample included 72,250 individuals aged 45 and above and their spouses across all states and union territories of India except Sikkim. The survey includes 31,464 older adults aged 60 and above, with 6749 oldest-old people aged 75 and above. The LASI adopts a multistage stratified area probability cluster sampling design to select the eventual units of observation. Households with at least one member aged 45 and above were taken as the eventual unit of observation [25]. This study provides scientific evidence on demographics, household economic status, chronic health conditions, symptom-based health conditions, functional and mental health, biomarkers, health care utilization, work and employment etc. It enables cross-state analyses and cross-national analyses of ageing, health, economic status and social behaviours and has been designed to evaluate the effect of changing policies and behavioural

outcomes in India. Detailed information on the sampling frame is available on the LASI wave-1 Report [25]. The effective sample size for the present study was 31,464 older adults in India.

2.2. Variable description

2.2.1. Outcome variable

The outcome variable “having diabetes” was binary in nature. The variable was assessed using the question “Has any health professional ever diagnosed you with diabetes?” The response was coded as 0 “no” and 1 “yes.” Such patient-reported outcome measures (PROM) do have their salience in this particular context [26].

2.2.2. Treatment variable

Frequent physical activity status was coded as yes (every day) and no (more than once a week, once a week, one to three times in a month and never). The question through which physical activity was assessed was “How often do you take part in sports or vigorous activities, such as running or jogging, swimming, going to a health centre or gym, cycling, or digging with a spade or shovel, heavy lifting, chopping, farm work, fast bicycling, cycling with loads?” [25].

2.2.3. Explanatory variables

Age was coded as young old (60–69 years), old-old (70–79 years) and oldest-old (80+ years). Sex was recoded as male and female. Educational status was categorized as no education/primary not completed, primary, secondary and higher. Marital status was coded as currently married, widowed and others. Others included divorced/separated/never married. Living arrangement was recorded as living alone, living with a spouse, living with children and others. Working status was categorized as never worked, currently working, currently not working and retired. Obesity/overweight was coded as no and yes. The respondents with a body mass index (BMI) of more than 25 were categorized as obese/overweight. Alcohol and tobacco consumption was recoded as no and yes.

Using household consumption data, the monthly per-capita consumption expenditure (MPCE) quintile was determined [25]. The sample households were canvassed using sets of 11 and 29 questions on food and non-food expenses, respectively. Food spending was gathered over a seven-day reference period, whereas the non-food expenditure was collected over 30-day and 365-day reference periods. The 30-day reference period has been used to standardise food and non-food expenses. The monthly per capita consumption expenditure (MPCE) is calculated and used to summarise consumption [25]. The variable was then divided into five quintiles, i.e., from poorest to richest. Religion was coded as Hindu, Muslim, Christian, and Others. Caste was recoded as Scheduled Tribe, Scheduled Caste, Other Backward Class, and others. The Scheduled Caste includes a group of the socially segregated population and financially/economically by their low status as per the Hindu caste hierarchy. The Scheduled Tribes (STs) and Scheduled Castes (SCs) are among the most disadvantaged and discriminated socio-economic groups in India. The OBC refers to persons who have been classified as “educationally, economically, and socially backward.” The OBCs are considered lower castes in the old caste system, although they are not outcastes. The “other” caste category is identified as having higher social status. Place of residence was coded as rural and urban. The regions of India were recoded as North, Central, East, Northeast, West, and South.

2.3. Statistical approach

Propensity score matching (PSM) analysis provides better estimates when one can retain a maximum number of factors that will

affect the treatment, but treatment should not affect the selected covariates. Table 1 contains a list of socio-economic and demographic variables such as household, demographic and individual variables. Stata 14 was used to conduct the statistical analysis [27].

2.3.1. Propensity score matching (PSM) analysis

PSM is a useful statistical method to estimate treatment effects for cross-sectional/observational/non-experimental data when randomized clinical trials are not possible [28]. The objective of this study is to make comparisons of outcomes between those older adults who engaged in frequent physical activity and those who did not engage. Finding the difference in outcome with and without treatment can be used to assess the effect of a treatment on an individual. If the selection of older adults who engage in frequent physical activity is random, the selection process is not correlated with the outcome of interest. In this instance, the average outcome for those older adults who engaged in frequent physical activity is compared with the average outcome for those who did not engage.

In cross-section surveys, treatment and control groups are not assigned at random, and those who were treated may differ in some way from those who were not. In this case, the occurrence of confounding variables may bias the estimation of the treatment effect (for example, frequent physical activity may be done only by older adults who meet certain requirements or belong to different socio-economic groups). The main interest for us is whether the difference observed in the prevalence of diabetes between those who received treatment and those who did not is attributable to the intervention or because older adults who receive treatment (frequent physical activity) belong to a different population.

As the number of variables to match rises, other matching approaches, such as frequency and pair-wise matched case-control designs that match outcomes for exposed and unexposed people with equivalent observable features, become increasingly difficult to employ. By allowing matching to be based on a score function of observable qualities, PSM helps to address this restriction [29].

Propensity Score is the probability that older adults will be treated or exposed to an intervention, given them various background characteristics [30].

$$p(X) = pr(D = 1|X)$$

Where $D = \{0, 1\}$, the indicator of exposure to treatment and X is the multidimensional vector of pre-treatment characteristics.

2.3.2. Counterfactual model

A counterfactual model has been constructed to calculate the average treatment (engaged in frequent physical activity) effect. Counterfactual is the potential outcome or the state of affairs that would have happened in the absence of the cause [31]. The Average Treatment Effect on the Treated (ATT) has been calculated using the counterfactual model. This metric assesses the treatment's impact on older adults who have been treated.

$$ATT = E(Y_1|D = 1) - E(Y_0|D = 1)$$

Where $(Y_1|D = 1)$ is the average outcome of the treated older adults. $(Y_0|D = 1)$ is the counterfactual, it depicts the average outcome that the treated individual would have achieved if treatment had not been given, which was not observed. The standard error (SE) of the difference of the average treatment effects was then calculated using the bootstrapping method and 100 replications [32].

Lastly, the Average Treatment Effect on the Untreated (ATU) older adults was measured, which shows the impact that the

Table 1

Socio-economic profile of older adults in India, LASI 2017-18.

Variables	Sample	Percentage
Diabetes		
No	26,906	85.5
Yes	4,475	14.2
Frequent physical activity		
No	25,792	82.0
Yes	5,672	18.0
Individual factors		
Age		
Young-old	18,408	58.5
Old-old	9,504	30.2
Oldest-old	3,552	11.3
Sex		
Male	14,931	47.5
Female	16,533	52.6
Education		
Not educated/primary not completed	21,334	67.8
Primary	3,532	11.2
Secondary	4,392	14.0
Higher	2,206	7.0
Marital status		
Currently married	19,453	61.8
Widowed	11,326	36.0
Others	685	2.2
Living arrangement		
Alone	1777	5.6
With spouse	6415	20.4
With children	21473	68.2
Others	1799	5.7
Working status		
Never worked	8,236	26.2
Currently working	9,435	30.0
Currently not working	11,483	36.5
Retired	2,301	7.3
Obesity		
No	25,324	80.5
Yes	6,140	19.5
Alcohol consumption		
No	26,882	85.4
Yes	4,582	14.6
Tobacco consumption		
No	18,903	60.1
Yes	12,561	39.9
Household factors		
MPCE^a quintile		
Poorest	6,826	21.7
Poorer	6,830	21.7
Middle	6,592	21.0
Richer	6,038	19.2
Richest	5,177	16.5
Religion		
Hindu	25,870	82.2
Muslim	3,550	11.3
Christian	899	2.9
Others	1,145	3.6
Caste		
Scheduled Caste	5,948	18.9
Scheduled Tribe	2,554	8.1
Other Backward Class	14,233	45.2
Others	8,728	27.7
Place of residence		
Rural	22,201	70.6
Urban	9,263	29.4
Region		
North	3,959	12.6
Central	6,598	21.0
East	7,442	23.7
Northeast	935	3.0
West	5,398	17.2
South	7,132	22.7
Total	31,464	100.0

^a MPCE: Monthly Per Capita Consumption Expenditure.

treatment would have had on those who did not participate

$$ATU = E(Y_1|D=0) - E(Y_0|D=0)$$

Where $E(Y_0|D=0)$ is the average observed outcome for the untreated older adults. $E(Y_1|D=0)$ is counterfactual, and it shows the average outcome that an individual would have obtained in the presence of treatment, which is unobserved.

Finally, we have performed a sensitivity analysis that determines how robustly unobserved confounding variables affect the selection into frequent physical activity to undermine the conclusions about true effects from a matching analysis. The details of the method are given elsewhere [28].

3. Results

Table 1 shows the sample characteristics of the study population. About 14% of older adults suffered from diabetes. Eighteen percent of older adults did the frequent physical activity. More than half of the older adults belonged to the young-old cohort, about 68% had no education/primary not completed, and 62% were currently married. Nearly six percent of older adults were living alone, about one-fourth were never working, and one-fifth were suffering from obesity. About 15% of older adults consumed alcohol and four in every ten were consumed tobacco. More than half of the older adults were Hindu, 19% belonged to scheduled caste, and a higher proportion of older adults living in rural areas.

3.1. Percentage of older adults who were engaged and not engaged in frequent physical activity (FPA)

The difference of treatment variable (frequent physical activity) by various background characteristics is shown in Table 2. The highest positive difference (treated-untreated) was observed among older adults who were currently working (45.6), male (21.2), belonged to young-old cohort (18.3), currently married (18.4), and those who consumed tobacco (10.9). Similarly, the highest negative difference was observed among older adults who were currently not working (−24.9), older females (−21.2), and those who were widowed (−18.9). The young-old cohort (73.6%) did more frequent physical activity than the old and oldest-old cohort. Moreover, frequent physical activity was more prevalent among older males than older female counterparts (64.9% vs 35.1%). Older adults who were currently working engaged more in frequent physical activity.

3.2. Results from propensity score matching analysis

Table 3 shows the results of propensity score matching analysis. The unmatched sample estimates for diabetes among older adults shows that the difference in the diabetes prevalence by those who frequently did the physical activity and those who did not do is −0.04 (or −4%). This indicates that older adults who did frequent physical activity were less likely to suffer from diabetes compared to those who did not do frequent physical activity. The estimated ATT values in treated and control groups were 0.123 and 0.147, respectively, indicating that the prevalence of diabetes among older adults was reduced by 2.5% points because of frequent physical activity. ATU results indicate that among older adults who did not do frequent physical activity if did frequent physical activity, the prevalence of diabetes is likely to decrease by 2.2% points.

After accounting for self-selection bias using PSM, the study observed a significant association of frequent physical activity and the prevalence of diabetes among older adults, supporting an indication of causality (Table 4).

3.3. Sensitivity analysis

Table S1 (supplementary file) shows the sensitivity analysis of obtained results using Mantel-Haenszel bounds and p-value for the average treatment effect on the treated older adults while setting the level of hidden bias to a certain value of C, i.e. 1, 1.1, 1.2 ... The $Q_{mh} +$ statistic adjusts the MH statistic downward for positive unobserved selection bias. Further, the table clearly indicates that the present result would be insensitive if around 19% bias is involved.

3.4. The overlap assumption

Numerous methods for testing the overlap assumption have been proposed in the previous literature, but the straightest forward is a visual analysis of the density distribution of the propensity score in the two groups [33]. According to the overlap assumption, each individual has a positive chance of receiving each treatment [27]. Estimated densities of the probability of getting each treatment were plotted to check whether the overlap assumption was violated. Fig. S1 (supplementary file) shows that the region where the estimated densities overlapped had a significant amount of mass. There is clear evidence that the overlap assumption was correct as there was a moderate amount of overlap between the two densities.

3.5. The covariate balance assumption

The density plot for the matched sample almost overlapped (Fig. S2 (supplementary file)), indicating that the covariates were balanced by matching the estimated propensity scores. The treatment assignment was strongly ignorable in the analysis because there two assumptions were met, implying that the estimated treatment effects were unbiased.

4. Discussion

The recent estimates suggest that the age-standardized DALYs for diabetes increased in India by 39.6% from 1990 to 2016, which was the highest increase among major non-communicable diseases [34]. Similarly, in the current population-based study of older Indian adults, there is a higher prevalence of diabetes, with 14.2% of the older participants reporting that they are diagnosed with diabetes. As documented, currently improved survival of patients with diabetes may have contributed to the increased rate of diabetes in older ages [35]. The study further found that frequent physical activity is inversely associated with the risk of developing diabetes mellitus, which is consistent with the findings of previous observational as well as experimental studies [36]. Regular physical activity, exercise and reduced total amount of sedentary time are recommended in previous studies in terms of better management of glycemic control and individuals' overall health [37]. Considering the higher risk of diabetic complications among South Asians linked to lower levels of physical activity [36], the current findings have major policy implications for the adoption and maintenance of physical activity to reduce the risk of diabetes mellitus.

The adoption and maintenance of moderate physical activity is critical in diabetic patients to manage their blood glucose levels and overall health [38]. Conversely, the sedentary lifestyle in old age is linked to their increased risk for diabetes and insulin resistance and related comorbidities [39]. Previous studies have also observed physical inactivity as an important component of developing geriatric syndrome, including dementia and falls in diabetic patients [40]. While treatment guidelines for diabetic people recommend that such individuals with sedentary lifestyles increase their

Table 2

Percentage of older adults who were engaged and not engaged in frequent physical activity by background characteristics in India, LASI 2017–18.

Variables	Not engaged in frequent physical activity	Engaged in frequent physical activity	Difference
Individual factors			
Age			
Young-old	55.2	73.6	18.3
Old-old	32.0	21.9	−10.1
Oldest-old	12.8	4.5	−8.3
Sex			
Male	43.6	64.9	21.2
Female	56.4	35.1	−21.2
Education			
Not educated/ primary not completed	68.8	64.1	−4.7
Primary	10.8	12.8	1.9
Secondary	13.7	14.7	1.0
Higher	6.6	8.5	1.8
Marital status			
Currently married	58.3	76.8	18.4
Widowed	39.6	20.7	−18.9
Others	2.1	2.5	0.4
Living arrangement			
Alone	5.9	4.6	−1.3
With spouse	19.6	23.5	3.8
With children	68.2	68.5	0.3
Others	6.2	3.4	−2.8
Working status			
Never worked	30.0	9.8	−20.2
Currently working	21.7	67.3	45.6
Currently not working	40.9	16.0	−24.9
Retired	7.3	6.9	−0.4
Obesity			
No	80.5	80.3	−0.2
Yes	19.5	19.7	0.2
Alcohol consumption			
No	86.8	80.0	−6.8
Yes	13.2	20.0	6.8
Tobacco consumption			
No	62.2	51.3	−10.9
Yes	37.8	48.7	10.9
Household factors			
MPCE* quintile			
Poorest	22.1	19.8	−2.3
Poorer	21.9	21.0	−0.9
Middle	20.5	23.1	2.6
Richer	18.9	20.5	1.6
Richest	16.6	15.6	−1.0
Religion			
Hindu	81.6	85.0	3.4
Muslim	11.8	9.0	−2.7
Christian	3.0	2.3	−0.7
Others	3.6	3.7	0.1
Caste			
Scheduled Caste	18.9	18.8	−0.2
Scheduled Tribe	7.7	10.0	2.3
Other Backward Class	44.6	48.0	3.3
Others	28.7	23.2	−5.5
Place of residence			
Rural	69.5	75.4	5.9
Urban	30.5	24.6	−5.9
Region			
North	13.3	9.3	−4.0
Central	21.5	18.3	−3.2
East	24.6	19.3	−5.3
Northeast	2.9	3.2	0.3
West	15.5	24.9	9.4
South	22.2	25.0	2.8
Total	100.0	100.0	0.0

Differences: Engaged in physical activity - not engaged in physical activity; *MPCE: Monthly Per Capita Consumption Expenditure.

Table 3

Results of matching estimates showing the effect of frequent physical activity on diabetes among older adults in India, LASI 2017–18.

Frequent physical activity	Treated	Control	Differences	SE	P > z
Diabetes					
Unmatched	0.123	0.162	−0.039	0.005	
ATT	0.123	0.147	−0.025	0.008	0.001
ATU	0.162	0.140	−0.022		
ATE			−0.022		

The model was controlled for all background characteristics.

SE: Standard error; ATT: Average treatment effect on the treated; ATU: Average treatment effect on the untreated; ATE: Average treatment effect.

physical activity [40], some studies have shown that diabetic older individuals who represent an even frailer population do less regular exercise than their non-diabetic counterparts [41].

A number of randomized controlled trials and prospective studies have examined the effects of exercise and diabetes-related interventions on changes in self-reported impairments in physical functioning and have shown significant improvements [42]. Some studies have shown physical activity as an important factor in preventing the onset of diabetes [43]. Again, a couple of studies have suggested that the treatment of diabetes involves physical activity and a healthy diet along with control of blood glucose levels and several therapies [44]. Similarly, moderate to high aerobic exercise is associated with substantially lower cardiovascular and overall mortality risks in diabetic patients [45]. Thus, considering the longitudinal studies conducted in different countries and populations that suggest that physical activity may decrease the influence of diabetes on the development of disability, improving glycemic control and weight loss, the programs focusing on the promotion of physical activity should be developed to ensure healthy ageing in Indian population. Recent studies in India also recommended promoting treatment-seeking and reducing the metabolic load to prevent diabetes in older ages [46].

Although previous studies have consistently reported an inverse association between physical activity and diabetes, most of these studies and the current study have focused on a structured leisure-time activity [42]. On the other hand, on-leisure-time activities, including activities related to housework and travelling, which are not considered in our study, were found to be major sources of daily energy expenditure in old age [47]. Moreover, the protective effects of physical activity may not only be related to vigorous fitness-enhancing activities but total energy expenditure as well [48]. Therefore, it is suggested to investigate the effects of physical activity on diabetes risk with a comprehensive evaluation of physical activities across all domains of daily life as an exposure indicator. Also, a regular evaluation of physical activities and their effects on diabetes in the follow-up population may clearly disclose the benefits of continuous physical activity on diabetes.

4.1. Limitation of the study

The major limitation of the present study is its cross-sectional design which restricts to assess the causality. However, the propensity-score matching approach provides a possible causal inference in the observed association by showing the treatment effects of physical activity on diabetes risk in the older Indian population. The self-reported nature of diabetes in the current study may also lead to several biases. As documented, nearly half of people in India with diabetes remain undetected, accounting for complications at the time of diagnosis [49]. This might have resulted in under-reporting, especially among socio-economically less privileged sections of the country [50]. Also, other factors of

Table 4
Adjusted regression coefficients (for diabetes) - unmatched and PSM models, LASI 2017-18.

Outcome variable (Diabetes)	Unmatched	PSM	
		ATE	ATT
Frequent physical activity (Yes vs no)	-0.03*(-0.04, -0.02)	-0.02*(-0.04, -0.01)	-0.03*(-0.04, -0.01)

The model was control for all background characteristics.

PSM: Propensity score matching; ATT: Average treatment effect on the treated; ATE: Average treatment effect.

reducing the risk of diabetes in individuals, such as eating habits and activities like yoga, are not considered in the present study. As evidence suggests, yogic exercises have enhanced the antioxidant defense mechanism in people with diabetes by reducing oxidative stress [51].

5. Conclusion

Our findings show that frequent physical activity is associated with a lower risk of diabetes in older adults. Given the fact that urbanization will be further accelerated and the ageing population will be an unprecedented challenge in India, our results underscore the urgent need to develop strategies of promoting an active lifestyle by maintaining physical activity to combat the diabetes epidemic, especially in the older population. However, since care and support in old age is made technology-based, and it has a positive impact on monitoring physical activity and fitness awareness through social media and other networks and diabetes self-management, these need to be further investigated with advanced research using advanced research comprehensive information on such factors.

Declaration of competing interest

The authors declare no conflict of interest.

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N/A.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.dsx.2021.102353>.

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